

Broken-shift overnight miss — root cause & fix plan

Client report: Harrison Richards · 24 Jun 2026 · Julie-Anne Seager inadequate rest · Verified against live ShiftCare data 14 Jul 2026

Verdict. Root cause is not the June 15 boundary fix. Overnight inadequate rest (<10h) that spans two local calendar days is excluded by a calendar-day gate in both import detection and pay-hours span collection. Julie-Anne's 19→20 Jun case is a confirmed false negative.

8.0h	False negative	2 files
Rest Julie had (need 10h)	Detector outcome today	Must change for fix

1. What the client saw

Harrison reported AI Studio failed to flag that Julie-Anne Seager (19 Jun) did not get the required 10-hour rest between shifts. OnCall moved to KC Studio as a workaround. Live ShiftCare (staff 1016147) confirms the inadequate rest sits across 19 Jun evening → 20 Jun morning — commonly described as “the 19th”.

2. Confirmed ShiftCare evidence

Date	ShiftCare ID	Start (AEST)	End (AEST)	Gap to next
Fri 19 Jun 2026	148393041	16:00	22:00	—
Sat 20 Jun 2026	148388885	06:00	14:00	8.0 h (BROKEN)
Sun 21 Jun 2026	156698330	07:30	14:00	17.5 h (OK)

Source: ShiftCare GET /users/areas.json?start=2026-06-15&end=2026-06-22 · filtered title Julie-Anne Seager

3. Why the issue came (timeline)

When	What	Effect
8 Apr 2026	Initial detect used gap < 10h/8h	Correct exclusive boundary
8 Jun 2026 (e3981f3)	Added cross-midnight day gate + flipped to gap > threshold	Good idea (cross-midnight), bad operator → false positives at exactly 10h
15 Jun 2026 (ebdcbc2)	Restored gap < threshold / gap ≥ threshold exit	Fixed exact-10h false positives; kept day gate unchanged
19–20 Jun 2026	Julie-Anne 22:00 → 06:00 (8h rest)	Day gate rejects overnight case → client-visible miss
24 Jun 2026	Harrison email + team leaves AI Studio	Trust damage; looks like “same bug came back”

Why it felt like a regression

Same SCHADS rest feature failed twice with opposite polarity: first too many flags (≤10h inclusive), then misses on true overnight shortfalls. Stakeholders experience both as “broken-shift validation is unreliable.”

4. Technical root cause

Functions: calculateIsBrokenShift / collectBrokenShiftSpanPrevious

Step 1 — gap check (correct after Jun 15). Broken only when $0 < \text{gap} < \text{threshold}$ (10h Personal Care / 8h sleepover).

Step 2 — calendar-day gate (the bug). Even if gap is short, return false unless sameStartDay OR spansOntoCurrentDay. Overnight 22:00 Day N → 06:00 Day N+1 has neither, so it is skipped.

Check	Julie 19→20 Jun	Passes?
gap (8h) < 10h	22:00 → 06:00	Yes
sameStartDay	starts 19 Jun vs starts 20 Jun	No
spansOntoCurrentDay	ends 19 Jun vs starts 20 Jun	No
Final isBrokenShift	gate fails	No — FALSE NEGATIVE

The day gate was added to support cross-midnight shifts that end early morning Day N with a follow-on later that same morning. That case needs covering — but not by excluding ordinary overnight shortfalls between two separate shifts.

5. How we solve it

Recommended fix (P0)

Treat inadequate rest as broken regardless of calendar day. After the gap threshold check succeeds, return true. Remove (or demote) the sameStartDay/spansOntoCurrentDay requirement for standard previous→next pairs. Clearer rule: any chronological previous→next pair with $0 < \text{gap} < \text{threshold}$ is broken.

Change	Where
Drop day-gate (or make gap-only)	shiftCsvParser.calculatesBrokenShift
Same change for OT/allowance spans	payHoursCalculator.collectBrokenShiftSpanPrevious
Add fixture: 22:00 Day1 → 06:00 Day2 = 8h → broken	payHoursCalculator.test.js (+ CSV/import tests)
Keep exact-10h NOT broken tests	Existing Jun 15 boundary tests stay
Re-upload timesheet + Compute Pay Hours	Ops — refresh persisted flags

Proposed rule (BR-BS)

Previous type	Broken when
Personal Care	$0 < \text{gap} < 10$ hours to next shift
Nursing Support	$0 < \text{gap} < 10$ hours
Sleepover	$0 < \text{gap} < 8$ hours
Exactly at threshold	Adequate rest — NOT broken
$\text{gap} \leq 0$ (overlap/contiguous)	Not broken via this rule

Test cases that must pass

Case	Expected
Julie pattern: Day N 22:00 → Day N+1 06:00 (8h)	Broken
8 PM → 6 AM next day (exactly 10h)	Not broken
9h 59m overnight	Broken
Same calendar day 2h gap	Broken
Sleepover predecessor, exactly 8h overnight	Not broken
Sleepover predecessor, 7.5h overnight	Broken

6. Related issues from same email

Issue	Status	Note
Vacant upload appends	Fixed 26 Jun (a53732d)	Stale rows soft-cancelled; confirm UI hides cancelled
48h vs 76h staff	Still open	ShiftCare Max fortnightly for Julie = 0 / not enforced; roster defaults 76; pay hard-codes 76

7. Rollout checklist

1. Patch both detection functions + tests (gap-only overnight).
2. Run backend suite; add Julie-Anne fixture from ShiftCare IDs above.
3. Deploy backend.
4. Re-import fortnight timesheet CSV covering 15–21 Jun; recompute pay hours.
5. Confirm exception report flags Julie-Anne 20 Jun shift as broken.
6. Reply to client with before/after on this named case.

Warning. Do not only “recompute” without code change — current code will keep missing overnight shortfalls by design. Existing Mongo may also lack the Jun 19–20 rows until re-import.

Evidence: live ShiftCare session · shifts 148393041 / 148388885 · code paths in shiftCsvParser.js + payHoursCalculator.js · prior boundary report scripts/output/broken-shift-boundary-regression-report.md